CN Practical 9: Install Wireshark and Capture Network Packets

Aim: To install Wireshark, configure it as a packet sniffer, and capture IP, TELNET and FTP packets.

Theory:

Wireshark is a network protocol analyser used to capture and examine packets travelling through a network interface. It is widely used for network troubleshooting, protocol analysis, security testing and educational purposes. It displays information such as source and destination addresses, protocol types, packet lengths and packet contents when available.

A packet sniffer captures network traffic visible to the selected interface. Wireshark supports display filters that allow users to examine specific protocols. The ip filter displays IPv4 packets, telnet filters TELNET traffic, and ftp filters FTP traffic. TELNET and traditional FTP are unencrypted protocols, so sensitive information may be exposed when they are used. They should be studied only in authorised lab environments.

Procedure:

1. Visit the official website https://www.wireshark.org/.
2. Download the installer compatible with the operating system.
3. Install Wireshark and the required packet-capture component.
4. Launch Wireshark after installation.
5. Select the active Ethernet or Wi-Fi network interface.
6. Start packet capture by double-clicking the interface.
7. Generate network traffic by visiting a website or performing an authorised network test.
8. Enter the display filter ip to view IPv4 packets.
9. Enter telnet to filter TELNET packets, if such traffic is available.
10. Enter ftp to filter FTP packets, if such traffic is available.
11. Select a packet to inspect its protocol layers and packet details.
12. Stop the capture and save it if required.

Observation:

Wireshark displayed packets captured from the selected network interface. The packet list included information such as source address, destination address, protocol and packet length. Display filters helped isolate specific protocol traffic. TELNET or FTP packets appeared only when corresponding traffic was generated and visible to the capturing interface.

Conclusion:

Wireshark was installed and used to capture and analyse network packets. The practical demonstrated packet capture, protocol filtering and inspection of network communication. It also highlighted the importance of using encrypted protocols for secure communication.